

MicroEMG

Help Guide

Step 1- Load MicroEMG recording and choose settings

MicroEMG Analysis GUI accepts Intan data files in .rhd format (single file per channel structure). Select the folder which contains all the signal files and load these into the program. Confirmation will be provided as to the length and number of channels of data selected.

Next, select any preconfigured settings files to use as default for the subsequent steps.

Step 2- Preprocessing

This interface is designed for preprocessing raw electromyography (EMG) data. Preprocessing removes noise and artefacts that can interfere with the interpretation of the data.

Key Features

- **Raw EMG:** Displays the original, unprocessed EMG signal.
- **Preprocessed EMG:** Displays the EMG signal after applying the selected preprocessing techniques.
- **Settings:** Allows users to customize the preprocessing parameters.

Preprocessing Settings

- **Remove mains noise (50 Hz):** This option removes the 50 Hz noise commonly found in electrical signals.
- **Filter:**
 - **Type:** Selects the type of filter to apply. The available options are:
 - **Bandpass:** Passes frequencies within a specific range and attenuates frequencies outside of that range.
 - **Lowpass:** Passes frequencies below a certain cutoff frequency and attenuates frequencies above that cutoff.
 - **Highpass:** Passes frequencies above a certain cutoff frequency and attenuates frequencies below that cutoff.
 - **Cutoff frequencies:** Sets the lower and upper cutoff frequencies for the filters.
 - **Order:** Specifies the order of the filter. Higher orders can provide better filtering but may introduce more phase distortion.

Usage Instructions

1. **Adjust settings:** Modify the preprocessing settings as needed based on the characteristics of the EMG data and the specific analysis goals.
2. **Apply preprocessing:** Click the "Apply" button to apply the selected preprocessing techniques to the raw EMG data.

3. **View results:** The preprocessed EMG signal will be displayed on the "Preprocessed EMG" tab.
4. **Adjust preprocessing settings:** If further filtering is needed, adjust the settings and reapply the filters.

Graph Controls

- **Amplitude scaling:**
 - **Up/down arrow:** Increases/decreases the signal amplitude (Up/down keyboard arrows)
 - The exact scaling factor may vary depending on the current amplitude range.
- **Navigation:**
 - **Single arrow keys:** Move through the data one division at a time.
 - **Double arrow keys:** Move through the data by a page at a time. (left/right keyboard arrows)
- **Time scale:**
 - **Divisions dropdown:** Selects the timebase over which signals are displayed. Options include milliseconds, seconds, and minutes.
 - **Slider:** Allows moving through the whole recording to a specific time. The slider displays the current position and total duration of the recording.

Step 3- Select Channels

This section of the interface allows you to select specific EMG channels for processing and analysis.

Key features:

- **Select all:** Selects all available channels.
- **Individual channel selection:** Click on the checkbox next to each channel number to select or deselect it.
- **Exclusion list:** The interface indicates which channels will be excluded based on the current selection.

Usage instructions:

1. **Select channels:**
 - If you want to process all channels, click the "Select all" button.
 - If you want to process only specific channels, click on the checkboxes next to the desired channel numbers.
 - The "Exclusion list" will show you which channels are not selected.

Step 4- Find motor units

Motor Unit Detection

This section of the interface allows you to detect motor units within the preprocessed EMG data.

Key features:

- **Detection sensitivity:** Controls the sensitivity of the motor unit detection algorithm.
- **Motor unit similarity:** Controls the similarity threshold for assigning potentials to the same motor unit.

Settings:

- **Detection sensitivity:**
 - Increasing the sensitivity threshold captures more motor units, but may also increase the number of false positives (i.e. noise transients that are incorrectly assigned to motor unit templates).
 - The default value is medium (0.1). You can adjust this value to suit your specific needs.
- **Motor unit similarity:**
 - Increasing the similarity threshold means that potentials assigned to the same motor unit template must be more similar in shape and amplitude, reducing in lower variation in motor unit templates, but also more action potentials being rejected.
 - The default value is medium (0.1). A higher value may result in fewer motor units being detected, but it can also help to reduce false positives.

Usage instructions:

1. **Adjust settings:** Modify the detection sensitivity and motor unit similarity settings as needed.
2. **Apply:** Click the "Apply" button to perform the motor unit detection.

Motor Unit Onset Times:

Once you apply the motor unit detection settings, the interface will display a graph showing the onset times of the detected motor units.

Graph explanation:

- **X-axis:** Represents time in minutes and seconds.
- **Y-axis:** Lists the detected motor units (MU 1, MU 2, etc.).
- **Vertical lines:** Indicate the onset times of motor unit potentials.

Interpretation:

- The graph visually represents the temporal distribution of motor unit activity within the EMG recording.
- You can use this graph to analyze the firing patterns of individual motor units and identify changes in muscle activity over time.
- For example, you might observe increased motor unit firing rates during periods of muscle contraction or decreased firing rates during periods of rest or relaxation.

Step 5- Select motor units

This section of the interface allows you to visualize the average potential of selected motor units.

Key features:

- **Motor unit selection:** Select one or more motor units for analysis.
- **Average potential:** Displays the average and individual motor unit potential waveform of the selected motor unit(s).
- **Channel selection:** Choose to view the motor unit potential on the triggering single channel only, or across all channels.

Additional notes:

- The graph shows the motor unit potential shape of the selected motor unit(s) across time.

- You can use this graph to analyze the characteristic shape of motor unit potentials and compare them across different motor units.
- The black "mean potential" line indicates the average amplitude of the waveform.
- The grey traces indicate the individual, superimposed EMG waveforms
- You can adjust the time scale and amplitude of the graph using the controls provided.

Step 6- Localise fibres

This section of the interface allows you to estimate the locations of individual muscle fibers within detected motor units.

Key features:

- **Time weighting:** Controls the influence of temporal information on fibre localization.
- **Fibre locations:** Displays the estimated locations of fibres within the selected motor unit(s).
- **Clustered fibre potentials:** Groups fibre potentials based on their estimated locations.

Settings:

- **Time weighting:**
 - Adjusts how much fibre potential timing (in addition to estimated locations) contributes to identifying different fibres.
 - A higher value gives more weight to temporal information, while a lower value gives more weight to estimated locations.

Usage instructions:

1. **Adjust time weighting:** Modify the time weighting setting as needed.
2. **Apply:** Click the "Apply" button to estimate fibre locations.

Additional notes:

- The accuracy of fibre localization depends on the quality of the EMG data and the chosen settings.
- The clustered fibre potentials is a 3D graph showing the fibre potential groupings and their relative onset times. It is possible for fibres to be co-existent in space, but have an offset in their inter-spike interval to enable separation.
- You can use the fibre localization results to study the spatial distribution of motor units within a muscle.

Step 7- Jitter analysis

This section of the interface allows you to analyze the jitter (variability in inter-spike intervals) of motor unit action potentials.

Key features:

- **Motor unit selection:** Select the motor unit to analyze.
- **Fibre pair selection:** Choose a pair of fibres to compare.
- **Mean consecutive differences:** Displays a heatmap showing the mean consecutive differences between the onset times of the two fibres.

- **Sample sizes:** Displays the number of consecutive differences for each fibre pair.
- **Raster plot:** Shows the onset times of individual fibre potentials and the inter-spike intervals.

Usage instructions:

1. **Select motor unit:** Choose the desired motor unit from the dropdown list.
2. **View results:** The heatmap and sample sizes will be displayed, showing the mean consecutive differences and the number of consecutive differences for each fibre pair.
3. **Select fibre pair:** Select a pair of fibres to analyze.
4. **View raster plot:** The raster plot will show the onset times of the two fibres and the inter-spike intervals between them.

Additional notes:

- The heatmap represents the average jitter between the two fibres, with darker colours indicating higher jitter and lighter colours indicating lower jitter.
- The sample sizes indicate the number of data points used to calculate the mean consecutive differences.
- The raster plot allows you to visualize the temporal relationship between the firing of the two fibres and to identify any patterns of asynchrony.

Step 8- Export results